

Task	Prompt	Answer
connected component number	The task is to determine the number of connected components in an undirected graph. A connected component is a subgraph where any two nodes are connected to each other by paths. Here is an undirected graph containing nodes from 1 to 10. The edges are: (1, 4), (1, 7), (1, 5), (1, 9), (1, 10), (1, 6), (1, 2), (4, 2), (4, 3), (4, 8), (4, 5), (4, 9), (4, 10), (7, 2), (7, 3), (7, 5), (7, 6), (7, 8), (7, 9), (5, 2), (5, 3), (5, 8), (5, 9), (5, 10), (9, 2), (9, 3), (9, 6), (9, 8), (9, 10), (10, 2), (10, 3), (10, 8), (6, 2), (6, 3), (6, 8), (2, 8), (2, 3). Question: How many connected components are there in the graph? Your answer should be an integer.	1
degree	The task is to determine the degree of a node in the graph. For the undirected graph, you should count the edge between two nodes only once. Here is an undirected graph containing nodes from 1 to 6. The edges are: (1, 6), (6, 5), (2, 3), (2, 4), (3, 5). Question: What is the degree of node 6 in the graph? Your answer should be an integer.	2
degree centrality	The task is to determine the degree centrality of a node in the graph. Degree centrality for a node is the fraction of nodes it is connected to. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 2), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (2, 6), (4, 3), (4, 5), (4, 7), (5, 3). Question: What is the degree centrality of node 3 in the graph? You need to format your answer as a float number.	0.5
density	The task is to determine the density of the graph. Density is defined as the ratio of the number of edges in the graph to the number of possible edges. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 2), (1, 3), (2, 3), (2, 4), (2, 5), (3, 4), (4, 5). Question: What is the density of the graph? You need to format your answer as a float number.	0.7
dfs	The task is to determine the depth-first search (DFS) traversal order given a starting node. Stop when the DFS cannot be continued. Here is an undirected graph containing nodes from 1 to 9. The edges are: (1, 2), (1, 3), (1, 6), (3, 9), (4, 8), (4, 5), (8, 7). Question: What is the depth-first search (DFS) traversal order for the starting node 1? You need to format your answer as a list of edges, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[(1, 2), (1, 3), (3, 9), (1, 6)]
diameter	The task is to determine the diameter of a graph. The diameter of a graph is the longest shortest path between any two nodes in the graph. The input graph is guaranteed to be connected. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 5), (1, 7), (1, 4), (5, 6), (2, 6), (2, 3). Question: What is the diameter of the graph? You need to format your answer as a float number.	5